

Algebra 2 [ADVANCED]

Unit 6: Solving Cube Root Equations

Day 2: Solving Practice

Name: _____

Key Concepts: $\sqrt[3]{x} = x^{\frac{1}{3}}$ • $(x^{\frac{1}{3}})^3 = x$ • $(x^{\frac{m}{n}})^{\frac{n}{m}} = x$ • **Even Numerator = $\pm$**
Perfect Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000

Part 1: Multi-Step Isolation

1. $3\sqrt[3]{x-4} + 2 = 11$

3. $3(x-2)^{1/3} = 18$

2. $2\sqrt[3]{2x-1} + 3 = 15$

4. $5 - 2\sqrt[3]{x} = -1$

Part 2: Radicals on Both Sides (Linear & Quadratic)

5. $2\sqrt[3]{2x-1} = \sqrt[3]{3x+5}$

7. $(15x^2 - 7x - 19)^{1/3} = (x^2 - 7x + 6)^{1/3}$

6. $\sqrt[3]{x^2 + 2x - 5} = \sqrt[3]{2x - 1}$

8. $(2x + 3)^{1/3} = (5x - 6)^{1/3}$

Part 3: Rational Exponents with Two Solutions ($\pm$)

9. $x^{2/3} = 16$

11. $(x+4)^{2/3} = 1$

10. $(x-1)^{2/3} = 4$

12. $(x-1)^{4/3} = 16$

Part 4: Advanced Polynomial Factoring

13. $\sqrt[3]{x^2 - 16x - 1} = -4$

15. $\sqrt[3]{7x^2 - 42x - 1} = (3x + 27)^{1/3}$

14. $\sqrt[3]{p^2 - 5p - 58} = -4$

16. $2(5p^2)^{1/3} = (8p + 144)^{1/3}$

Part 5: Challenge — Variable Expressions & Non-Factorable Results

17. $\sqrt[3]{3x + 8} = x + 2$

19. $(x^2 - 2x + 1)^{1/3} = (2x - 1)^{1/3}$

18. $\sqrt[3]{4x} = x$

20. $\sqrt[3]{9x^2 - 7x + 2} = \sqrt[3]{-2x + 12}$

Part 6: Real-World Applications

21. **Flow Rate:** Radius (r) and flow rate (Q) are related by $\sqrt[3]{r} = 0.5Q$. If $Q = 10$, find r .

22. **Metabolism:** Torso length (L) and mass (M) follow $\sqrt[3]{L} = 0.1M$. If $M = 15$, find L .

Solve

1. $\sqrt[3]{x^2 - 1} = 2$

3. $x^{7/8} = 128$

2. $(x + 10)^{2/3} = 25$

4. $\sqrt[3]{x^2 - 7x - 10} = 2$

Answer Bank

-135	-64
-7	-6
-5	-3
-2	-0.6
0	1
1.5	2
3	3.4
6	7
8	9
27	31
64	115
125	218
1000	256
108.5	$\pm \frac{5\sqrt{14}}{14}$
-8	1.4

Reflection

Compare what happens to the sign of a number when it is cubed versus when it is squared. Use this to explain why $x^{2/3} = k$ has two solutions, but $x^{1/3} = k$ has only one.